Alternative options: code-made graphics

Seven graphic types for the film. Each figure has a number; the explanation with the same number follows it. Content in the figures is sample content for design only.

Common rules for all five

- These graphics are made by code, never drawn or generated pictures (owner rule). They show real data and stay allowed after the no-generated-pictures rule.
- TIMING IS NOT THE JOB OF THESE GRAPHICS. The main visual planner decides if a graphic is used, where, for how long, how often and in which order. A graphic only receives from the planner: its data, duration_s, and a list of cues (time offsets for each reveal). It animates inside that duration, with all animation lengths as fractions of it. No graphic has its own minimum or maximum seconds, interval or share in its config.
- Interface (same for all seven): can_render(data) returns ok or a plain reason; required_cues(data) returns how many reveal steps there are, so the planner can plan; render(data, duration_s, cues, language) returns the frames. If the planner gives fewer cues, the graphic says which steps it will skip; it never invents timing.
- Fallback order for a beat without real material (existing rule): another crop of a real picture, open-data map / aerial / street, real document / quote / timeline / board, longer hold, report the shortage. Never black, never a drawing. These graphics belong to the third step. How long a still may be held stays in the planner's coverage config.
- Output: 1920×1080 frames (design 960×540 at 2x), 16:9. A 9:16 variant for shorts and reels is a later step; keep text inside safe areas now.
- Render with the method the app already uses for code-made graphics (browser rendering as in the zoom maps). Deterministic: same data, same cues, same frames.
- Fonts are stored in the repo, not loaded from the internet at render time. Mockup fonts: Barlow Condensed (titles), Newsreader (text), IBM Plex Mono (labels), all open licences; Persian: Vazirmatn; Arabic: Noto Naskh Arabic (already in the repo fonts folder). Check licences and add the Latin fonts the same way.
- Languages en, de, fa, ar. Persian is always in Persian script. For fa and ar the whole layout is mirrored. Text comes from the translated narration and fact labels through the existing language pipeline, never translated at render time.
- Sample content in the mockups (bracketed quotes, times, dates, names) is only for design. A graphic is never rendered with placeholder text: if data is missing, can_render says so and the planner uses the next fallback.
- Spoiler firewall and sensitive content: nothing shows the outcome before the narration does. No victim imagery, no graphic content, no minors. Names only as the case dossier allows.
- Credit: every graphic with an outside source shows it on screen (map provider, document source). Rights and editorial sign-off is not self-certified by the agent (existing rule).
- No new paid calls. Numbers live in config/parameters, model names never in code. Tests first. No dead code. English UI. Schema change only by a new migration (probably none).

Order of implementation (loop)

- Step 0: read the existing modules for code-made graphics (zoom maps, detective board, document and text fill ways, coverage and fill_ways config, the visual planner) and list what can be reused. Do not duplicate.
- Step 1: shared renderer: fonts in the repo, theme tokens (colours, type scale), RTL mirroring, the interface can_render / required_cues / render, credit footer.
- Step 2: the graphics in this order of value: 4 Document card, 6 Conflict page, 7 Conflict board, 1 Timeline, 2 Map zoom, 5 Recap board, 3 Floor plan (needs a real plan source).

- Step 3: planner integration: each graphic registers as a shot type with its data requirements. The planner (not the graphic) decides use, position, duration and cues. The conflict board state is kept in the film plan.
- Step 4: tests (golden images per language, cue accuracy, spoiler, verbatim quote, no placeholder rendering, board state, mirrored layout). Live proof on case 9 in the real app: draft film with these graphics, report seconds and % per kind and beats still without material.
- Step 5: audit, docs (one doc for these graphics), small commits after the secret scan, push, check CI once.

1 Timeline



Figure 1: Timeline (16:9 sample, 1920×1080)

1 Timeline: explanation and implementation

PURPOSE

Shows the order of events as a line with dated nodes. Real data, no pictures.

OFFER TO THE PLANNER

Offered to the planner when the chapter has 3 or more dated facts. The planner decides if, when and for how long.

INPUTS

`events[]`	date (day, month or year precision), label (max 70 characters), fact_id
`cues[]`	one cue per node reveal, with its time offset, set by the planner
`duration_s`	set by the planner
`language`	fa and ar mirror the whole layout

LAYOUT

- Canvas 1920×1080 (design 960×540 at 2x). Background #0F1318, soft radial light top left.
- Header: small mono label (amber) and a 58 px condensed title. Line at 55% height, 816 px wide, 2 px grey; the filled part is amber, 4 px.
- 3 to 5 nodes visible. Nodes alternate above and below the line (first above). Node: big date (condensed, amber), year (mono, grey), one sentence (serif 18 px, max 3 lines).
- Current node has a halo. Nodes after the current one are not drawn; the next one may appear as an empty ring only when its cue arrives (spoiler firewall).
- More than 5 events: the line pans so the current node stays between 40% and 60% of the width.

MOTION (DRIVEN BY PLANNER CUES)

- The graphic reveals node k when cue k arrives. Between cues nothing moves except the halo.
- All animation times are fractions of the given duration, never fixed seconds. If duration_s is too short for the data, the graphic reports it to the planner and does not render.

DATA AND SAFETY RULES

- Events come only from stored case facts (fact_id kept). Dates are shown exactly as the fact states them; no day if the fact has none.
- Never invent or round an event. An event that reveals the outcome stays hidden until its cue.

CONFIG KEYS (CONFIG/PARAMETERS)

- `timeline.max_nodes_visible = 5`
- `timeline.node_text_max_chars = 70`
- `timeline.pan_target_range = [0.4, 0.6]`

ACCEPTANCE TESTS

- Golden image per language (en, de, fa, ar) with fixed sample data and fixed cues.
- No hidden event appears in any frame before its cue (test over all frames).
- Longest allowed label does not overflow; fa and ar are mirrored and in the correct script.

2 Map zoom

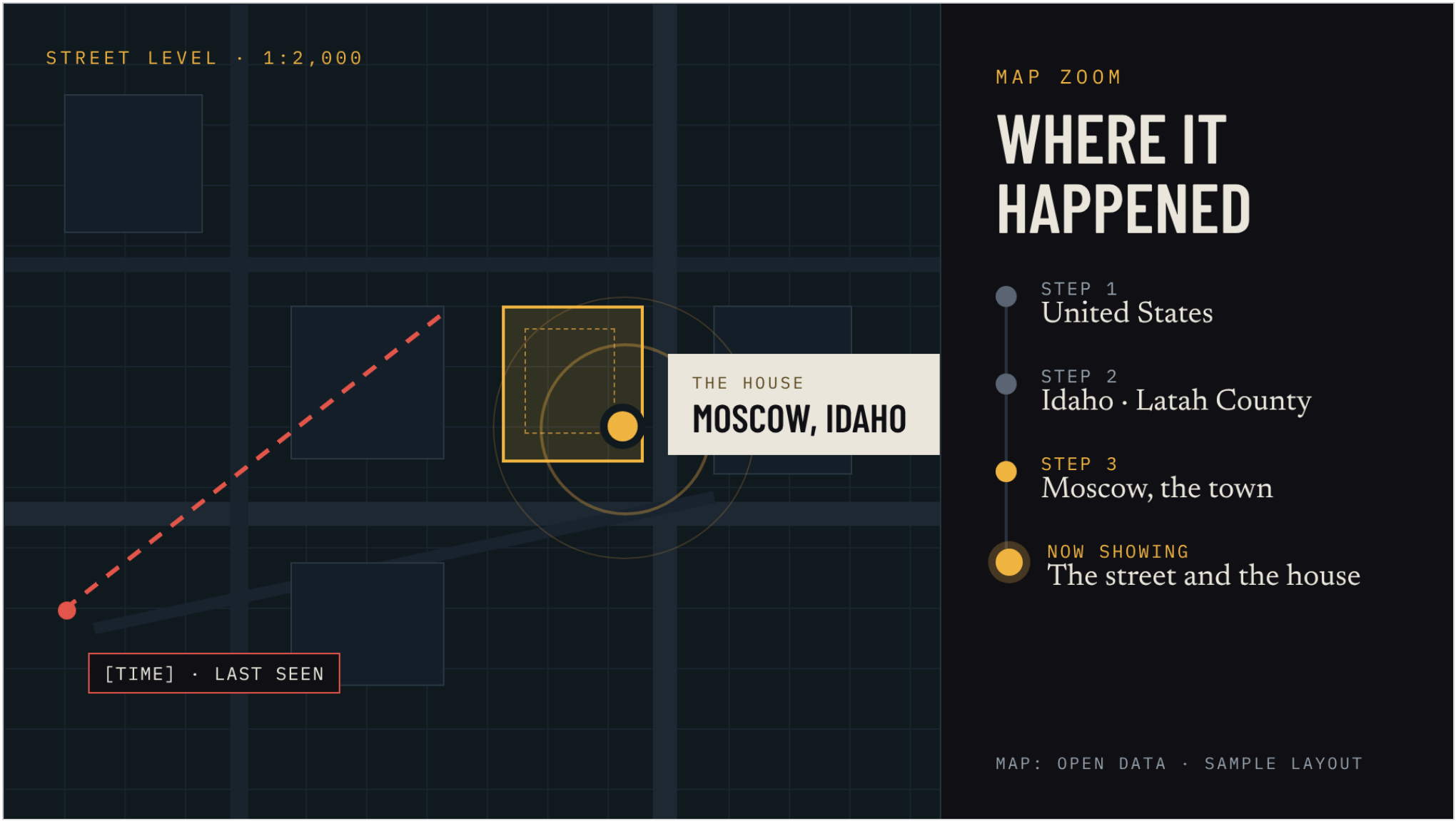


Figure 2: Map zoom (16:9 sample, 1920×1080)

2 Map zoom: explanation and implementation

PURPOSE

Shows where something happened: zoom from country to town to street to house, with a route when the story moves.

OFFER TO THE PLANNER

Offered to the planner when the beat names a located place or a route. Extends the zoom maps the app already has.

INPUTS

`place`	name, coordinates (from the app's geocoder), zoom steps (country, region, town, street)
`route[]`	optional points with a time label taken from a fact (fact_id)
`marker`	kind (house / scene / last seen) and label
`cues[] and duration_s`	set by the planner; one cue per zoom step and for the route

LAYOUT

- Left 65%: the map. It must be a REAL open-data map or aerial image from the app's providers, never a drawn city. The grid in the mockup is only a style sample.
- Marker: amber dot with two thin rings; target building outlined amber. Label card (paper colour, dark text) with the place name.
- Route: dashed red line, round start point, mono chip with the time (from a fact only).
- Right 35%: panel with the zoom steps as a vertical list; current step amber with halo.
- Footer: map source and licence (from config per provider).

MOTION (DRIVEN BY PLANNER CUES)

- Each zoom step starts on its cue. The zoom is fitted to the duration the planner gives (slot-fitted), never stretched past it, so picture and narration cannot drift.
- If fewer cues than steps arrive, the graphic skips the upper steps and starts at the nearest level; the planner is told which steps were used.

DATA AND SAFETY RULES

- Coordinates only from the geocoder or stored case facts. If the place cannot be located with confidence, the graphic does not render and says why.
- Private addresses of people who are not part of the public record: show street or block, not the house.

CONFIG KEYS (CONFIG/PARAMETERS)

- `map_zoom.steps = [country, region, town, street]`
- `map_zoom.credit_by_provider = {...}`

ACCEPTANCE TESTS

- Every step starts exactly on its cue (tolerance one frame).
- Credit visible on every map frame.
- No map without a located place; the reason is returned to the planner.

3 Floor plan

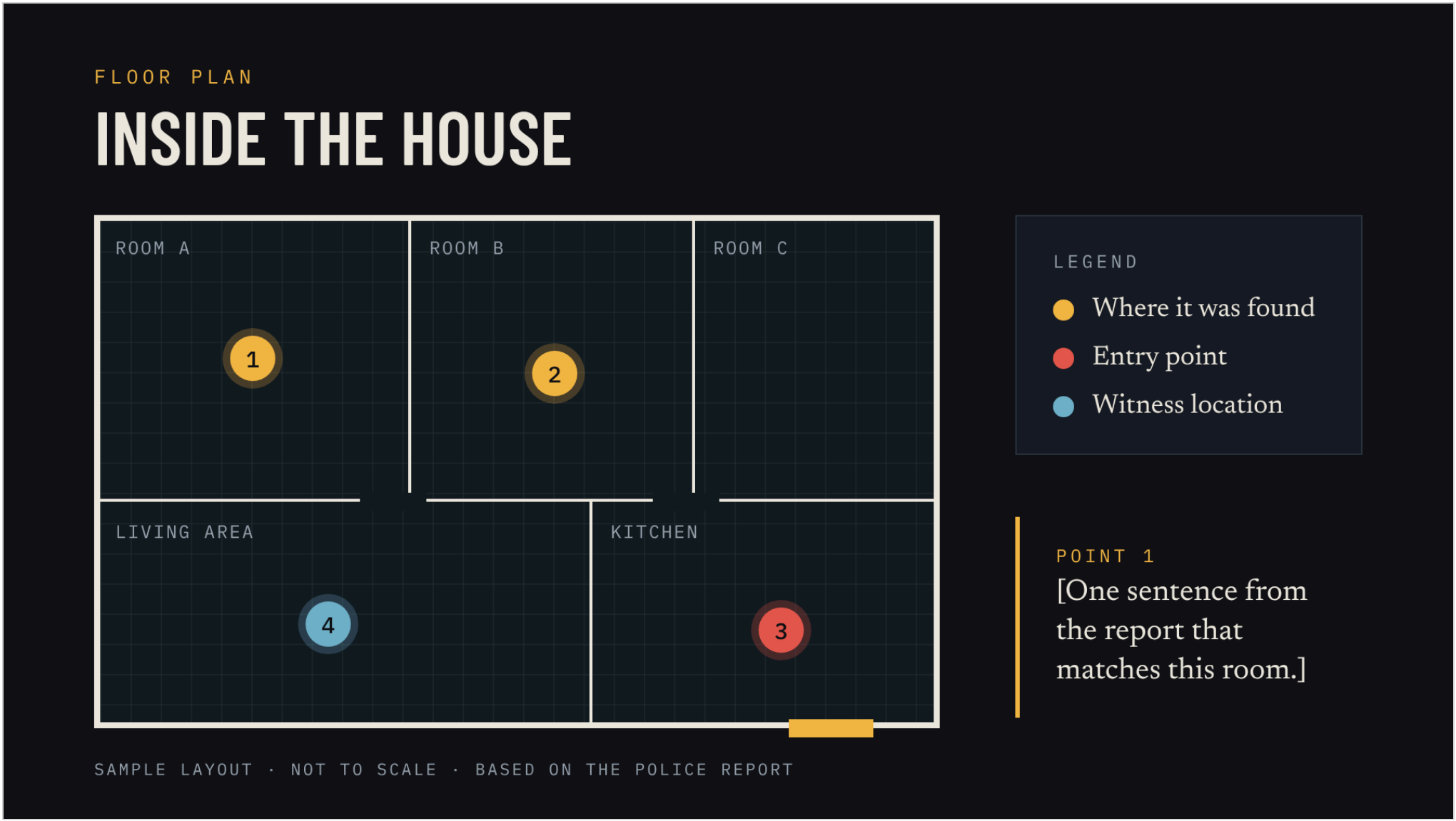


Figure 3: Floor plan (16:9 sample, 1920×1080)

3 Floor plan: explanation and implementation

PURPOSE

Explains where things happened inside a building, room by room, with numbered markers.

OFFER TO THE PLANNER

Offered to the planner only when a real plan exists in the case sources (police report, court exhibit, public listing).

INPUTS

`rooms[]`	id, label, rectangle on a grid, from the source plan
`markers[]`	number, kind (found / entry / witness), room id, short text, fact_id
`source`	document id and page of the plan
`cues[] and duration_s`	set by the planner; one cue per marker

LAYOUT

- Plan area left (about 58%), legend and a note card right. Thick outer wall, thin inner walls, door gaps, entry point in amber.
- Markers: numbered circles with a soft ring. Amber = found, red = entry, blue = witness. Colour is never the only signal: number and legend explain it.
- Note card shows the sentence of the current marker (amber rule on the left).
- Footer: 'schematic, not to scale' and the source of the plan.

MOTION (DRIVEN BY PLANNER CUES)

- Walls and rooms are drawn first (a fixed share of the duration), then marker k appears on cue k. A reveal never runs ahead of its cue.

DATA AND SAFETY RULES

- No real plan: do NOT invent a layout. Allowed fallback: labelled boxes made only from facts that name rooms and floors, footer 'schematic from the report, not a plan'.
- No marker is drawn as a body or position of a victim. Markers are neutral numbers (graphic-content rule).

CONFIG KEYS (CONFIG/PARAMETERS)

- `floor_plan.max_markers = 8`
- `floor_plan.legend_kinds = [found, entry, witness]`

ACCEPTANCE TESTS

- A case without a plan source yields either the labelled schematic or nothing (test both).
- Marker order equals cue order; legend colours pass a contrast check.
- fa and ar: mirrored layout, labels in the right script.

4 Document card

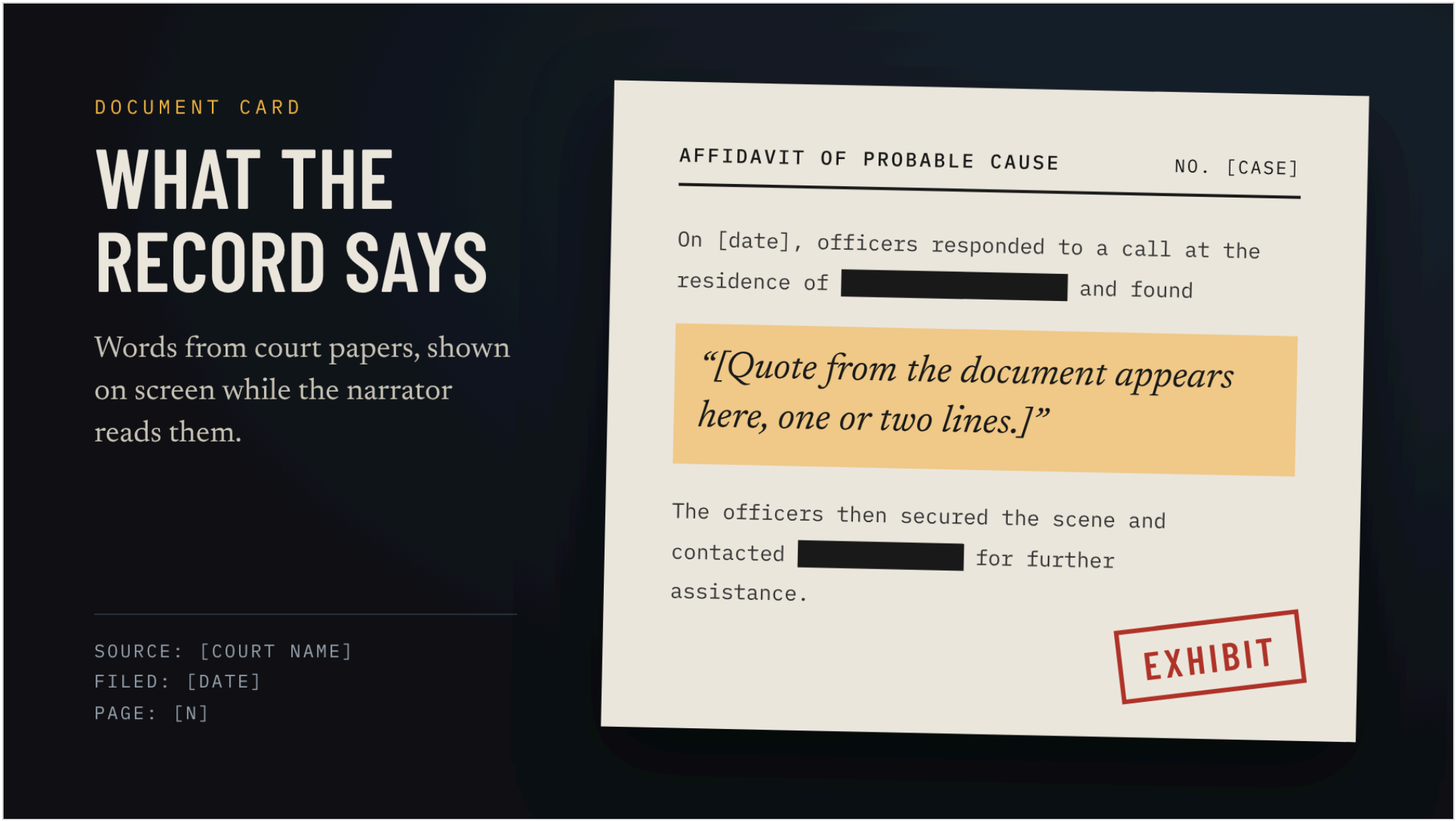


Figure 4: Document card (16:9 sample, 1920×1080)

4 Document card: explanation and implementation

PURPOSE

Puts the exact words of a court or police document on screen while the narrator reads them.

OFFER TO THE PLANNER

Offered to the planner when the beat quotes a stored document. Gives the planner something real for a beat without a picture.

INPUTS

`document`	type, court or agency, case number, date, page, stored document id, source URL
`quote`	verbatim text, one or two lines (about 160 characters), its character span in the stored document, fact_id
`context lines`	real lines before and after the quote from the same document, optional
`redactions[]`	spans the source itself redacts
`cues[] and duration_s`	set by the planner; cue for the quote highlight

LAYOUT

- Left column: label, large title 'What the record says', one short sentence, source block (source, filed date, page) in mono.
- Right: paper card rotated about 1 degree, shadow, header line with document type and case number.
- Quote line highlighted in amber, serif italic 25 px. Context lines in mono, grey. Redaction bars black.
- Stamp 'EXHIBIT' only if the document is an exhibit in the record.

MOTION (DRIVEN BY PLANNER CUES)

- Card enters at the start of the given duration. The highlight wipes across the quote starting at its cue and ending at the planner's end cue.

DATA AND SAFETY RULES

- Text must be verbatim from the stored document. NEVER paraphrase, cut inside a quote, or invent filler lines (the mockup uses placeholders only to show the design).
- If a picture of the real page exists, prefer a crop of the real page with the quote highlighted over a typeset card.
- Names the source redacts stay redacted; minors and protected data follow the existing rules.

CONFIG KEYS (CONFIG/PARAMETERS)

- `document_card.quote_max_chars = 160`
- `document_card.rotate_degrees = 1.0`
- `document_card.stamp_only_if_exhibit = true`

ACCEPTANCE TESTS

- A test compares the quote with the stored document text: must match character for character.
- No card without a stored source id.
- Golden images for en, de, fa, ar. For fa and ar the quote stays in the document's original language; the narration is translated (decide per case, log it).

5 Recap board



Figure 5: Recap board (16:9 sample, 1920×1080)

5 Recap board: explanation and implementation

PURPOSE

Sums up what is known, what is disputed and what is still open. Helps long films.

OFFER TO THE PLANNER

Offered to the planner at chapter ends. How often, and when, is the planner's decision.

INPUTS

`confirmed[]`	facts established by court findings or at least two independent sources, each with fact_id (max 3)
`disputed[]`	claims contested by a party, with who contests, linked to conflict ids (max 2)
`unknown[]`	open questions from the story blueprint (max 2)
`progress`	elapsed film seconds and total film seconds, from the planner
`cues[] and duration_s`	set by the planner; one cue per card

LAYOUT

- Title top left. Three equal cards: Confirmed (amber rule), Disputed (blue), Still unknown (red), each 260 px wide, items in serif 20 px, max 60 characters.
- Bottom: progress bar with elapsed time left, total right, marker at the current position.
- Colours always come with a text label.

MOTION (DRIVEN BY PLANNER CUES)

- Card k appears on cue k. The marker on the bar moves in with the cards.

DATA AND SAFETY RULES

- Only facts already narrated before this point (spoiler firewall). Never show the outcome early.
- Wording comes from the fact ledger; it may be shortened but not changed in meaning. The agent logs the basis for each 'confirmed' item.

CONFIG KEYS (CONFIG/PARAMETERS)

- `recap.max_items = {confirmed: 3, disputed: 2, unknown: 2}`

ACCEPTANCE TESTS

- No item refers to an event after the current film second (test).
- Every item traces to a fact id; an item without one is not shown.
- fa and ar mirrored; no overflow.

6 Conflict page

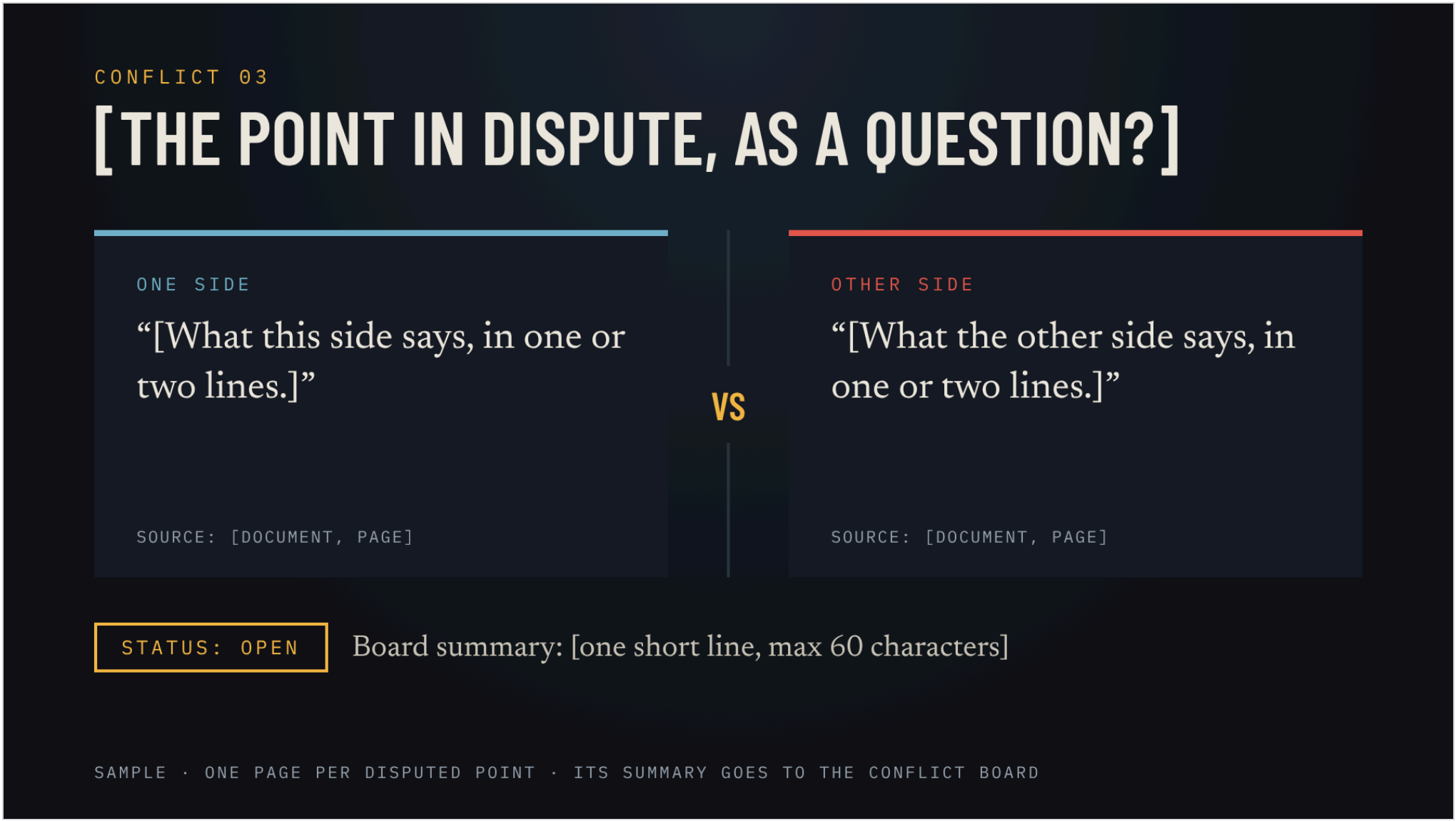


Figure 6: Conflict page (16:9 sample, 1920×1080)

6 Conflict page: explanation and implementation

PURPOSE

One page for one disputed point in the case: what one side says, what the other side says, the source of each, and the status.

OFFER TO THE PLANNER

Offered to the planner for every conflict the case dossier lists. Two kinds: between parties (prosecution against defense) and between witnesses or evidence. The page says who contradicts whom.

INPUTS

`conflict`	id, number, title as a question (max 70 characters), kind (parties / witnesses / evidence), status (open / decided / refuted)
`sides[2]`	who says it (party or witness), claim text (max 160 characters, a quote or a close paraphrase marked as such), source document and page, fact_id
`summary`	one line for the board (max 60 characters)
`cues[] and duration_s`	set by the planner; cue per side and for the status

LAYOUT

- Header: 'CONFLICT 03' (mono, amber) and the question as a large condensed title (max 2 lines).
- Two cards of equal size with a thin 'VS' divider. Left rule blue, right rule red. Each card: side label (who), claim in serif 25 px, source line in mono at the bottom.
- Below: status pill (OPEN amber, DECIDED blue, REFUTED grey) and the board summary line.
- Side colours are only accents: the labels name the sides, so meaning never depends on colour.

MOTION (DRIVEN BY PLANNER CUES)

- Left card on cue 1, right card on cue 2, status pill on cue 3. The planner may omit the status cue while the point is open.

DATA AND SAFETY RULES

- Both sides come from the fact ledger with sources. Never invent a side and never take sides: the page shows claims, not the app's judgement.
- Status 'decided' or 'refuted' only after the narration has said so (spoiler firewall). Before that the page shows 'open'.
- Claims that quote a document use the same verbatim rule as the document card.

CONFIG KEYS (CONFIG/PARAMETERS)

- `conflict.claim_max_chars = 160`
- `conflict.title_max_chars = 70`
- `conflict.summary_max_chars = 60`
- `conflict.kinds = [parties, witnesses, evidence]`

ACCEPTANCE TESTS

- Every claim has a source and a fact id; without them the page is not rendered.
- Status never ahead of the narration (test).
- fa and ar mirrored; long claims shrink within a limit and are then cut by the data rule, never by the renderer.

7 Conflict board



Figure 7: Conflict board (16:9 sample, 1920×1080)

7 Conflict board: explanation and implementation

PURPOSE

The collection of all conflict summaries. It starts empty and grows while the film runs: a card is added after its conflict page has been shown.

OFFER TO THE PLANNER

Offered to the planner after each conflict page, at recaps and at the end. The planner decides when it is shown and for how long.

INPUTS

`cards[]`	conflict id, number, title (short), side A line, side B line, status; only conflicts whose page the planner has already used
`total`	number of conflicts in the case, shown only if the planner allows it (it reveals the film's structure)
`new_card`	id of the card to animate in
`cues[] and duration_s`	set by the planner

LAYOUT

- Header: 'CONFLICT BOARD' and 'What is still disputed'; counter 'N of M told' top right (optional).
- Grid of 3 columns, up to 2 rows per view (6 cards); more cards: the board scrolls or splits into a second view chosen by the planner. Each card: number, status pill, short title, one line per side.
- Slots for conflicts not yet told are dashed and labelled 'not yet told' only if the planner allows it; otherwise they are hidden.

MOTION (DRIVEN BY PLANNER CUES)

- The new card enters on its cue; older cards stay still. A status change animates the pill only, on its cue.

DATA AND SAFETY RULES

- The board state is part of the film plan: it is built from the order in which the planner used the conflict pages, never from the dossier alone, so it can not show anything the viewer has not seen.
- Each card is the summary line of its conflict page, same wording.

CONFIG KEYS (CONFIG/PARAMETERS)

- `conflict_board.columns = 3`
- `conflict_board.max_cards_per_view = 6`
- `conflict_board.show_total = false`

ACCEPTANCE TESTS

- The board never contains a conflict whose page has not been used earlier in the film plan (test over all frames).
- Wording of every card equals the summary of its page.
- Handles 1 to 12 cards; fa and ar mirrored.